

Tab 1

AICTE Cyber Security Internship – Technical Progress Report (2025–2026)

1. Intern Information

- Intern Name: Shaik Mohammed Omar
 - Institute Name: Keshav Memorial Institute of Technology
 - State: Telangana
 - Programme / Specialization: B.Tech, 2nd year, CSE.
 - Team Name (SIH): Team ASHTOJ
 - Email ID: god.s.m.o.97@gmail.com
-

2. Project Information

- SIH Problem Statement Code: SIH25161
 - Problem Statement Title: Mitigating National Security Risks Posed by Large Language Models (LLMs) in AI-Driven Malign Information Operations
-

3. Mentor / SPOC Details

- Assigned Mentor Name: Shri Ojas Dedhiya
 - Mentor / SPOC Email: Ojasdedhiya@gmail.com
-

4. Project Implementation Summary

A. Implementation Progress Summary:

During this reporting period, the primary focus was on conducting a structured literature survey and producing a formal Literature Review document for TattvaDrishti — the SIH 2025 prototype platform for detecting and mitigating LLM-enabled malign information operations.

The literature review systematically surveyed peer-reviewed research across five core technical domains underpinning the TattvaDrishti architecture: AI-generated text detection (RoBERTa-based classifiers, DetectGPT, watermarking), stylometric analysis, semantic risk scoring via local LLMs, graph-based threat intelligence, and federated intelligence sharing (STIX/TAXII). This documentation work is a prerequisite for the enterprise-grade hardening and deployment phase of the internship, providing the academic and technical grounding required for formal integration into AICTE's cyber security architecture.

B. Development & Integration Details:

TattvaDrishti implements a composite risk scoring architecture with the following weighted signal pipeline:

- Ollama Local LLM Semantic Scoring (40%) — contextual risk assessment using locally hosted LLMs (llama3.2:3b / mistral), ensuring data sovereignty for sensitive content
- HuggingFace RoBERTa Classifier (35%) — AI-generated content detection via roberta-base-openai-detector
- Behavioural Analysis Module (15%) — scoring urgency markers, authority claims, and manipulation tactics
- Stylometric Analysis Module (10%) — linguistic fingerprinting via vocabulary richness and sentence-level features

Supporting infrastructure includes: Fernet symmetric encryption and Ed25519 asymmetric signatures for federated sharing packages, STIX/TAXII-aligned sharing scaffold, Server-Sent Events (SSE) for real-time analyst notifications, SQLite persistence layer, and a Next.js analyst dashboard.

Documentation produced this week maps each component to its foundational research basis, which is a required step before enterprise integration and SOC workflow alignment.

C. Weekly Technical Update:

- Conducted comprehensive literature survey covering 18 peer-reviewed works across AI detection, stylometry, LLM-based semantic scoring, graph intelligence, and federated sharing

- Produced formal Literature Review document (9 sections, ~3,500 words) grounding TattvaDrishti's architecture in published research
- Developed comparative analysis table benchmarking TattvaDrishti against prior approaches
- Identified research gaps addressed by the system: integration gap, real-time gap, and federated sharing gap
- Document serves as technical foundation for the upcoming enterprise hardening and AICTE SOC integration phase

5. Technical Challenges / Constraints

A. Technical Challenges / Constraints:

- Ollama local inference requires sufficient hardware (RAM $\geq$ 8GB) for the llama3.2:3b model; performance on lower-spec machines is constrained, which impacts real-time scoring latency
- The RoBERTa-based detector is susceptible to paraphrase-based evasion attacks, as documented in the literature (Krishna et al., 2023) — motivating the multi-signal ensemble approach
- Python 3.11.x dependency constraint (due to FastAPI + Pydantic v1 compatibility) limits deployment flexibility on environments running Python 3.12+

B. Operational / Integration Issues:

- Federated ledger is currently a lightweight simulation; full STIX/TAXII schema implementation and blockchain-backed sharing receipts are planned for a subsequent phase
- SQLite is used for local persistence; migration to a production-grade database will be required for enterprise-scale deployment within AICTE's infrastructure
- NDA and formal onboarding documentation are pending receipt from AICTE — access to AICTE internal systems and SOC workflows has not yet been provisioned, which has limited integration work to the prototype level

6. Planned Technical Activities & Next Steps

A. Planned Technical Activities:

- Begin enterprise hardening of the TattvaDrishti backend — input validation, rate limiting, authentication layer, and logging for SOC-grade deployment

- Explore migration from SQLite to PostgreSQL for scalable persistence
- Draft integration proposal for aligning TattvaDrishti's detection pipeline with AICTE's SOC incident response playbooks
- Evaluate PEFT/LoRA fine-tuning of the HuggingFace classifier on domain-specific disinformation datasets

B. Upcoming Development Tasks:

- Implement authentication and role-based access control (RBAC) for the analyst dashboard
- Add automated remediation hooks triggered on high composite risk scores
- Extend federated sharing module toward fuller STIX/TAXII schema compliance
- Produce API documentation and deployment runbook for AICTE infrastructure team

7. Implementation Status

- Current Status: In Progress
- Attendance Status: Regular
- Reporting Week No.: Week-1
- Date of Submission: 8th June, 2026.

8. Mentor / SPOC Observation & Remarks

Mentor / SPOC Signature:

Date:

Tab 2

AICTE Cyber Security Internship – Technical Progress Report (2025–2026)

1. Intern Information

- Intern Name: Shoaib Sadiq Salehmohamed
 - Institute Name: Keshav Memorial Institute of Technology
 - State: Telangana
 - Programme / Specialization: B.Tech, 2nd year, CSE.
 - Team Name (SIH): Team ASHTOJ
 - Email ID: shoaibkulsums@gmail.com
-

2. Project Information

- SIH Problem Statement Code: SIH25161
 - Problem Statement Title: Mitigating National Security Risks Posed by Large Language Models (LLMs) in AI-Driven Malign Information Operations
-

3. Mentor / SPOC Details

- Assigned Mentor Name: Shri Ojas Dedhiya
 - Mentor / SPOC Email: Ojasdedhiya@gmail.com
-

4. Project Implementation Summary

A. Implementation Progress Summary:

During this reporting period, the primary focus was on conducting a structured literature survey and producing a formal Literature Review document for TattvaDrishti — the SIH 2025 prototype platform for detecting and mitigating LLM-enabled malign information operations.

The literature review systematically surveyed peer-reviewed research across five core technical domains underpinning the TattvaDrishti architecture: AI-generated text detection (RoBERTa-based classifiers, DetectGPT, watermarking), stylometric analysis, semantic risk scoring via local LLMs, graph-based threat intelligence, and federated intelligence sharing (STIX/TAXII). This documentation work is a prerequisite for the enterprise-grade hardening and deployment phase of the internship, providing the academic and technical grounding required for formal integration into AICTE's cyber security architecture.

B. Development & Integration Details:

TattvaDrishti implements a composite risk scoring architecture with the following weighted signal pipeline:

- Ollama Local LLM Semantic Scoring (40%) — contextual risk assessment using locally hosted LLMs (llama3.2:3b / mistral), ensuring data sovereignty for sensitive content
- HuggingFace RoBERTa Classifier (35%) — AI-generated content detection via roberta-base-openai-detector
- Behavioural Analysis Module (15%) — scoring urgency markers, authority claims, and manipulation tactics
- Stylometric Analysis Module (10%) — linguistic fingerprinting via vocabulary richness and sentence-level features

Supporting infrastructure includes: Fernet symmetric encryption and Ed25519 asymmetric signatures for federated sharing packages, STIX/TAXII-aligned sharing scaffold, Server-Sent Events (SSE) for real-time analyst notifications, SQLite persistence layer, and a Next.js analyst dashboard.

Documentation produced this week maps each component to its foundational research basis, which is a required step before enterprise integration and SOC workflow alignment.

C. Weekly Technical Update:

- Conducted comprehensive literature survey covering 18 peer-reviewed works across AI detection, stylometry, LLM-based semantic scoring, graph intelligence, and federated sharing

- Produced formal Literature Review document (9 sections, ~3,500 words) grounding TattvaDrishti's architecture in published research
- Developed comparative analysis table benchmarking TattvaDrishti against prior approaches
- Identified research gaps addressed by the system: integration gap, real-time gap, and federated sharing gap
- Document serves as technical foundation for the upcoming enterprise hardening and AICTE SOC integration phase

5. Technical Challenges / Constraints

A. Technical Challenges / Constraints:

- Ollama local inference requires sufficient hardware (RAM $\geq$ 8GB) for the llama3.2:3b model; performance on lower-spec machines is constrained, which impacts real-time scoring latency
- The RoBERTa-based detector is susceptible to paraphrase-based evasion attacks, as documented in the literature (Krishna et al., 2023) — motivating the multi-signal ensemble approach
- Python 3.11.x dependency constraint (due to FastAPI + Pydantic v1 compatibility) limits deployment flexibility on environments running Python 3.12+

B. Operational / Integration Issues:

- Federated ledger is currently a lightweight simulation; full STIX/TAXII schema implementation and blockchain-backed sharing receipts are planned for a subsequent phase
- SQLite is used for local persistence; migration to a production-grade database will be required for enterprise-scale deployment within AICTE's infrastructure
- NDA and formal onboarding documentation are pending receipt from AICTE — access to AICTE internal systems and SOC workflows has not yet been provisioned, which has limited integration work to the prototype level

6. Planned Technical Activities & Next Steps

A. Planned Technical Activities:

- Begin enterprise hardening of the TattvaDrishti backend — input validation, rate limiting, authentication layer, and logging for SOC-grade deployment

- Explore migration from SQLite to PostgreSQL for scalable persistence
- Draft integration proposal for aligning TattvaDrishti's detection pipeline with AICTE's SOC incident response playbooks
- Evaluate PEFT/LoRA fine-tuning of the HuggingFace classifier on domain-specific disinformation datasets

B. Upcoming Development Tasks:

- Implement authentication and role-based access control (RBAC) for the analyst dashboard
- Add automated remediation hooks triggered on high composite risk scores
- Extend federated sharing module toward fuller STIX/TAXII schema compliance
- Produce API documentation and deployment runbook for AICTE infrastructure team

7. Implementation Status

- Current Status: In Progress
- Attendance Status: Regular
- Reporting Week No.: Week-1
- Date of Submission: 8th June, 2026.

8. Mentor / SPOC Observation & Remarks

Mentor / SPOC Signature:

Date:

MOM

Date: 14th June, 2026.

Time: 8:40 - 9:10 pm

Turn it into a metric

How to sell this, success criteria

Define the goals: Defining what to do, then Plan

1. Draft the problem statement, aligning with the goal
2. Literature review in depth, with what's currently there in the literature review. (imp, backing to sell our project)
3. Define 5ws 1h about the solution.
4. Clear goals, at the end vision. Roadmap. What we need to do.
5. Review at least 25 to 30 papers for Literature review. Download papers and read 10-15 papers
6. Talk to 15-20 ppl for research papers, understand the process

Write down goals and metrics we want to achieve in this prototype etc

1. Basic goal is to have a working prototype of this
2. Define 5W and 1H
3. Draft up a comprehensive problem statement / goal statement of it
4. Literature survey of current market availability etc, also market available products, research, current available tech in it etc.
5. What do we think of doing in this internship
6. Estimates of what all tasks we will do etc, how much coding, when will we start to work on testing etc, how much to research etc,
7. Making roadmap to identify stuff how and what will be able to do etc
8. We will understand what our actual goal
9. Go through atleast 20-25 papers etc
10. Missed some points
11. Go and speak to atleast 10-15 ppl and understand how to find papers etc etc. send cold messaging to phd's etc on linkedin.
12. Try to

Date: 23th June, 2026.

Is there a practical way to scan all of social media and give results.

Instead of doing this what if we take input from users instead.

What if we search about certain people who are flagged like on radar and like search based on the people detecting

Do a very comprehensive feasibility analysis of doing everything. Do it very efficiency

Doing a full feasibility testing of all

Deployment + platform is not yet to be decided

Feasibility analysis of the whole current pipeline + diagrams + flowchart of the current plan

Setup jira too

//

algo to filter the info out,

search for the dataset thats trending

feasibility analysis of the current pipeline +

build usp

deployment platform

map your thoughts; visio. diagram flowchart; project workflow +

jira create for project; start adding work and goals

5th July:

1. Simplify the info for the users, should be understandable to the users, target the SMEs
2. Simplify the user guided dashboard
3. Go to police officer!? (ask them to use it, feedback); Research; (psychologist); 4 ppl
4. He wants to see the change in the prototype. /////
5. Do a literature survey build more, comprehensive
6. Updating the Jira board, with all the past works.
7. Do not miss weekends, ping Ojas sir for the weekend. ////
8. Prepare an explanation deck and pitch for your product.
9. Proof of concept stage, dk our computation yet whole, so do it later. Find end target user
10. Cold calling phd, teachers on LinkedIn
11. Brainstorm and refine the scope

14th July:

1. Scan instagram posts reels, make it into an input pipelines for different social media platforms. Telegram / reddit. (Telegram)
2. Improve our models
3. Analyze group messages
4. Demonstrate why our product is better (Create a simulation how others are doing it and how we are doing it). Live demonstration // start this week not finish it this week
5. This weekend, show the demo. // This Weekend . Sunday evening 9p.m.

19th july:

1. Fix UI of the mapping thing
2. ex we have suspect, and then wanna scan for one person, soo can we take all activity from multiple sources etc, soo can we screen for all activities
3. take auto location instead of manual location only
4. simplify and explain + visualise the graph
5. TLS 3.0 encryption use this instead of this of blockchain. lets replace the blockchain instead of this, use encryption.
6. why do we use this product, also how.
7. simulation of the product and its use. - work on the simulation part of things.

30th July:

1. work on the vid tmr
2. meet with prashant sir(saturday)
3. How the authorities can regulate the law, cjp protest switch to our product and how we are addressing the gaps (video)

Week-1

AICTE Cyber Security Internship

Team Progress Report — Team ASHTOJ

Submitted for Mentor Review — Stipend Processing Cycle

Team Name	ASHTOJ
Institute	Keshav Memorial Institute of Technology (KMIT), Hyderabad
Problem Statement Code	SIH25161
Problem Statement Title	Mitigating National Security Risks Posed by Large Language Models (LLMs) in AI-Driven Malign Information Operations
Interns	Shaik Mohammed Omar, Shoaib Sadiq Salehmoahmed
Assigned Mentor	Shri Ojas Dedhiya
College SPOC	Ms Asha Sheldon

1. WORK COMPLETED BY THE TEAM

During this reporting period, the team focused on consolidating the research and academic foundation of TattvaDrishti — the SIH 2025 winning prototype for detecting and mitigating LLM-enabled malign information operations — in preparation for the enterprise hardening phase of the internship.

Jointly, the team produced a comprehensive Literature Review document grounding TattvaDrishti's multi-signal detection architecture (AI-content classification, stylometric analysis, semantic risk scoring, graph intelligence, and federated sharing) in 18 peer-reviewed research works. This review also includes a comparative analysis benchmarking TattvaDrishti against existing approaches in the field, and a forward-looking roadmap outlining the technical phases for enterprise-grade deployment within AICTE's infrastructure.

2. INDIVIDUAL CONTRIBUTION

Shaik Mohammed Omar

- Studied research on AI-generated text detection, semantic risk scoring through local LLMs, and federated intelligence sharing, and worked through how each of these applies to TattvaDrishti's existing pipeline.
- Went through TattvaDrishti's architecture component by component (Ollama semantic scoring, HuggingFace classifier, behavioural and stylometric modules) and traced each one back to the research it's grounded in, to make sure the design choices actually hold up.
- Worked out a phase-wise plan for the rest of the internship — sequencing enterprise hardening, SOC integration, and deployment milestones based on what's realistically achievable in the remaining time.
- Put together and finalised the literature review and supporting documentation submitted this week.

Shoaib Sadiq Salehmoahmed

- Looked into existing AI-content detection and disinformation-mitigation tools in the market to understand where TattvaDrishti stands relative to current solutions.
- Read through academic papers in the disinformation detection and stylometric analysis space and pulled out the parts relevant to the review.
- Went back through the cited sources to check they were technically accurate and actually relevant to TattvaDrishti's detection pipeline before they were included.
- Fed findings on how the project compares to existing tools into the comparative analysis section of the review.

3. NEXT STEPS

- Begin enterprise hardening of the TattvaDrishti backend (authentication, input validation, logging) per the roadmap drafted this week.
- Initiate alignment discussions for SOC workflow and incident-response playbook integration with AICTE's cyber security architecture.

Mentor / SPOC Signature

A handwritten signature in blue ink that reads "Asha.m". The signature is written in a cursive, flowing style. The "A" is large and loops around, and the "m" has a long, sweeping tail.

Week-2

AICTE Cyber Security Internship

Technical Progress Report (2025–2026)

All India Council for Technical Education — E-Governance & Cyber Security Cell

1. INTERN INFORMATION

Intern Name	Shaik Mohammed Omar, Shoaib Sadiq Salehmohamed
Institute Name	Keshav Memorial Institute of Technology
State	Telangana
Programme / Specialization	B.Tech, 2nd Year, CSE
Team Name (SIH)	Team ASHTOJ
Email ID	god.s.m.o.97@gmail.com , shoaibkulsums@gmail.com

2. PROJECT INFORMATION

SIH Problem Statement Code	SIH25161
Problem Statement Title	Mitigating National Security Risks Posed by Large Language Models (LLMs) in AI-Driven Malign Information Operations

3. MENTOR / SPOC DETAILS

Assigned Mentor Name	Shri Ojas Dedhiya
Mentor / SPOC Email	Ojasdedhiya@gmail.com

4. PROJECT IMPLEMENTATION SUMMARY

A. Implementation Progress Summary

This week's focus shifted from documentation to internship-level planning — converting last week's literature review into a structured, metric-driven roadmap for the remainder of the engagement. A dedicated planning session was held to define success criteria, scope the problem statement more precisely, and lay out a clear sequence of goals for the coming weeks.

The session produced a working definition of the project's 5W1H (What, Why, Who, When, Where, How), a refined problem statement aligned to the original SIH goal, and a phased roadmap connecting research depth to eventual prototype hardening. Alongside this, outreach was initiated

toward identifying and sourcing additional academic literature, including direct contact with a few researchers to better understand how to locate and evaluate relevant papers.

B. Development & Integration Details

No code-level changes were made this week; the priority was establishing a sound research and planning foundation before further hardening work on TattvaDrishti's pipeline (Ollama semantic scoring, HuggingFace classifier, behavioural and stylometric modules) resumes.

The roadmap drafted this week explicitly sequences the upcoming technical work: deepening the literature base, finalising the problem statement and success metrics, and only then proceeding to enterprise hardening and SOC integration tasks identified in the Week 1 report.

C. Weekly Technical Update

Week 2 (09 June – 14 June 2026):

- Held a structured planning session to define project goals, success metrics, and a working roadmap for the internship
- Drafted a refined problem statement aligned with the original SIH goal
- Defined the solution's 5W1H to clarify scope and direction
- Set a target of reviewing 25–30 literature review papers, with 10–15 to be read in depth; began identifying and downloading candidate papers
- Began outreach to researchers and peers (target: 10–15 contacts) to learn how to source and evaluate relevant literature; initial contact made with a few individuals
- Outlined estimates for the internship's remaining work — split across research, coding, and testing phases — to inform the next two weeks of execution

5. TECHNICAL CHALLENGES / CONSTRAINTS

A. Technical Challenges / Constraints

- Sourcing high-quality, relevant papers within the target range (25–30) is proving time-intensive, particularly for niche sub-topics like federated threat intelligence sharing
- Determining the right balance between research depth and development time was a key planning challenge this week, since over-indexing on literature could delay the hardening phase

B. Operational / Integration Issues

- Outreach to researchers/PhDs for guidance on literature sourcing is still in early stages; response rate so far has been limited
- NDA and formal onboarding documentation from AICTE are still pending, which continues to limit work to the prototype level rather than AICTE infrastructure

6. PLANNED TECHNICAL ACTIVITIES & NEXT STEPS

A. Planned Technical Activities

- Continue reviewing the targeted 25–30 papers, with focus on reading 10–15 in depth over the coming week
- Finalise the problem statement and success metrics drafted in this week's planning session
- Continue outreach to researchers/PhDs (target: 10–15 total contacts) for guidance on the literature review process
- Translate the roadmap's first phase into concrete technical tasks for resuming TattvaDrishti development

B. Upcoming Development Tasks

- Begin enterprise hardening of the TattvaDrishti backend per the roadmap (carried over from Week 1: authentication, input validation, logging)
- Reassess SQLite-to-PostgreSQL migration timeline based on finalised roadmap
- Revisit the SOC integration proposal in light of the refined problem statement and goals

7. IMPLEMENTATION STATUS

Current Status:

- ☐ Not Started
- ☒ In Progress
- ☐ Completed
- ☐ Blocked

Attendance Status:

- ☒ Regular
- ☐ Irregular
- ☐ On Leave

Reporting Week No.

Week 2

Date of Submission	15 June 2026
--------------------	--------------

8. MENTOR / SPOC OBSERVATION & REMARKS

--

Mentor / SPOC Signature	
-------------------------	--

Week-3

AICTE Cyber Security Internship

Team Progress Report — Week 3

Team ASHTOJ | KMIT, Hyderabad | SIH25161

Team Name	ASHTOJ
Institute	Keshav Memorial Institute of Technology (KMIT), Hyderabad
Problem Statement Code	SIH25161
Problem Statement Title	Mitigating National Security Risks Posed by Large Language Models (LLMs) in AI-Driven Malign Information Operations
Interns	Shaik Mohammed Omar, Shoaib Sadiq Salehmoahmed
Assigned Mentor	Shri Ojas Dedhiya
College SPOC	Ms Asha Sheldon
Reporting Week	Week 3 (23 June 2026)

1. Work Completed by the Team

Week 3 was focused on thinking through how TattvaDrishti actually scales beyond a single-input prototype — asking hard questions about whether scanning social media at volume is practical, what the realistic entry points into the system are, and what the architecture needs to look like before any further development begins.

The central question the team sat with this week was whether building a system that continuously scans social media is the right approach at all — and what the alternatives are. Two alternative input models were evaluated: a user-report model (where users submit content they suspect is AI-generated or misleading) and a person-of-interest model (where the system monitors content associated with flagged individuals or accounts). Both were discussed as potentially more operationally precise than broad social media sweeping.

The main deliverables this week were: a comprehensive feasibility analysis of the full pipeline and its batch/multi-platform extension, a system architecture document with a detailed flowchart, and the initial setup of a Jira project board for tracking internship tasks going forward.

2. Individual Contributions

Shaik Mohammed Omar

- Drove the core feasibility analysis, working through each component of the current pipeline (FastAPI, Ollama, HuggingFace, behavioural and stylometric modules, SQLite) to assess how each would behave under batch load, where bottlenecks sit, and what the development effort looks like to make each stage batch-capable.
- Mapped out the full system architecture and produced a detailed flowchart covering the end-to-end pipeline from analyst intake and multi-signal detection through to SSE broadcast, federated sharing, and ledger sync to give a clear picture of the current system before any changes are made.

- Contributed to the Jira board setup, adding tasks related to the research and dataset sourcing work planned for upcoming weeks.
- Worked through the three input models (automated social media scan, user-submitted content, flagged person-of-interest monitoring) and thought through the trade-offs of each in terms of precision, feasibility, and alignment with AICTE's use case.

Shoaib Sadiq Salehmohamed

- Looked into what datasets and trending content sources are currently available that could feed into TattvaDrishti — both public datasets and platform-level content streams — to assess what the system would actually have to work with in practice.
- Researched algorithms and filtering approaches for pulling out relevant or high-risk content from a noisy social media feed, given that not everything ingested would need full pipeline analysis.
- Helped think through the person-of-interest monitoring model — what it would take to build a system that tracks content around flagged accounts or individuals rather than scanning broadly.
- Set up the Jira project board and began structuring it with tasks, goals, and milestones for the remaining internship weeks.
- Created a flow of how all the components work in the application using visio.

3. Deliverables Produced This Week

- Feasibility Analysis of Current Pipeline and Possibility of Multi-Platform Batch Ingestion — a full technical document covering pipeline component analysis, platform-by-platform API assessment (Twitter/X, Reddit, Telegram, Facebook), proposed batch architecture, identified gaps, and a phased development roadmap
- System Architecture Document — includes the basic system flowchart for TattvaDrishti covering all stages from intake to federated ledger sync, with the orchestrator, detection engines, provenance checks, graph intelligence, and SSE stream all mapped out
- Jira Project Board — initial setup complete with tasks and goals for the internship

4. Key Decisions and Open Questions

Input Model

The team is not yet settled on whether broad social media scanning is the right primary input model. The user-submission model and the flagged-account monitoring model were both identified as potentially more operationally targeted. The recommendation is to keep the current intake endpoint as the user-submission interface and add the flagged-account model as a second input mode, with automated social media sweeping as a longer-term goal rather than the immediate priority.

Ollama as the Bottleneck

The feasibility analysis confirmed that Ollama local LLM inference is the single biggest constraint on any batch processing extension. Running it synchronously on 50+ items would take 2.5–12.5 minutes. The proposed fix — a semaphore-limited async worker pool plus a fast-path that skips Ollama for items below a HuggingFace confidence threshold — was agreed on as the right mitigation before any batch mode is built.

Deployment Platform

Deployment platform and hosting environment are not yet decided. This was flagged as an open item. The feasibility analysis notes that SQLite is sufficient for prototype-scale work but migration to PostgreSQL will be required before enterprise deployment. The decision on deployment environment will inform which database and infrastructure choices make sense.

5. Planned Activities — Week 3

- Finalise the problem statement and define success metrics and USP for TattvaDrishti in the context of AICTE's cyber security architecture
- Begin implementing Phase 1 of the batch roadmap: HuggingFace batched inference, asyncio parallelisation for behavioural and stylometric modules, SQLite WAL mode
- Continue literature review — target 25–30 papers, with 10–15 read in depth; outreach to researchers ongoing
- Decide on deployment platform and document the target environment
- Continue populating Jira with tasks for the remaining internship weeks

Mentor / SPOC Signature

A handwritten signature in blue ink, reading "Asha.M", is displayed on a light gray rectangular background.

Week-4

AICTE Cyber Security Internship

Team Progress Report — Week 4

Team ASHTOJ | KMIT, Hyderabad | SIH25161

Team Name	ASHTOJ
Institute	Keshav Memorial Institute of Technology (KMIT), Hyderabad
Problem Statement Code	SIH25161
Problem Statement Title	Mitigating National Security Risks Posed by Large Language Models (LLMs) in AI-Driven Malign Information Operations
Interns	Shaik Mohammed Omar, Shoaib Sadiq Salehmohamed
Assigned Mentor	Shri Ojas Dedhiya
College SPOC	Ms Asha Sheldon
Reporting Week	Week 4 (30 June 2026)

1. Work Completed by the Team

Week 4 shifted focus from architecture and feasibility planning toward usability and validation. Having mapped the full pipeline in Week 3, the team turned to how the system would actually be used and understood by non-technical stakeholders, while also stepping back to conduct a comprehensive review of the current application with the mentor's input.

A full walkthrough of TattvaDrishti's existing features was carried out with Shri Ojas Dedhiya, covering the pipeline end to end and identifying which parts of the current build genuinely serve the end user versus which are primarily internal/technical scaffolding. This review directly informed the decision to prioritise simplifying the dashboard for subject-matter-expert (SME) users, since the current interface surfaces a level of technical detail (model scores, pipeline stage logs, raw signal outputs) that is not meaningful to a non-technical analyst or reviewer.

The team also began identifying who the realistic end user of TattvaDrishti actually is, recognising that this had not been concretely defined despite the architecture work done so far. This is treated as a proof-of-concept-stage question — full computational scale is deliberately being deferred until the end-user profile and dashboard experience are validated first.

2. Individual Contributions

Shaik Mohammed Omar

- Led the simplification effort on the user-facing dashboard, reviewing current UI elements against the mentor's feedback and identifying which technical outputs need to be abstracted or hidden for SME users.
- Participated in the comprehensive feature review of the current application alongside the mentor, walking through each existing module to assess relevance and clarity from an end-user perspective.
- Prepared the current prototype for a mentor walkthrough, in response to the mentor's request to see the changes made to the prototype directly.
- Contributed to Jira board clean-up, adding historical tasks and tagging outstanding items for Week 5 planning.
- Contributed to early discussion on defining the realistic end target user for TattvaDrishti, ahead of deeper outreach planned for Week 5.

Shoaib Sadiq Salehmoahmed

- Supported the feature review by cataloguing existing pipeline outputs and flagging which fields are too technical for non-expert users.

- Began outreach groundwork by cold-calling and messaging PhD researchers and academic contacts on LinkedIn to gather external perspective on realistic end users and use cases.
- Took part in the brainstorming session to refine project scope in light of the mentor's feedback.
- Updated the Jira board to reflect all past work completed across Weeks 1–3, backfilling tasks and closing out completed items so the board accurately tracks project history going forward.

3. Deliverables Produced This Week

- Feature Review Notes — a consolidated set of observations from the mentor-led walkthrough of the current application, flagging which outputs need simplification for SME users
- Prototype Walkthrough — an updated build presented to the mentor showing progress since Week 3
- Updated Jira Board — backfilled with all past work from Weeks 1–3 and restructured to track upcoming priorities
- Preliminary notes on candidate end-user profiles, informed by early outreach conversations

4. Key Decisions and Open Questions

Dashboard Simplification

The mentor's review confirmed that the current dashboard is too technical for its intended SME audience. The team has agreed to prioritise simplifying the user-facing dashboard before any further backend or scale work, so that outputs are understandable to a non-technical reviewer rather than requiring pipeline-level knowledge to interpret.

Deferring Full-Scale Computation

The team explicitly agreed to treat the project as being at proof-of-concept stage. Full-scale computation and batch performance work (as scoped in the Week 3 feasibility analysis) is being deliberately deferred until the end target user is clearly defined and the simplified dashboard has been validated.

Defining the End Target User

Who the system is ultimately for is still an open question. Outreach to PhD researchers and academics on LinkedIn was initiated this week specifically to help narrow this down, alongside internal brainstorming to refine scope.

5. Planned Activities

- Continue simplifying the user-guided dashboard based on mentor feedback from this week's walkthrough
- Present the updated prototype to the mentor and gather further feedback on the simplified interface
- Continue populating and refining the Jira board with tasks for upcoming weeks
- Continue cold outreach to PhD researchers and academics on LinkedIn to help define the end target user
- Hold a follow-up brainstorming session to further refine project scope based on outreach findings and mentor input
- Revisit full computation/scale planning once the end user and dashboard direction are clearer

Mentor / SPOC Signature

Asham

Tab 6

AICTE Cyber Security Internship

Technical Progress Report (2025–2026)

All India Council for Technical Education — E-Governance & Cyber Security Cell

1. INTERN INFORMATION

Intern Name	Shoaib Sadiq Salehmohamed
Institute Name	Keshav Memorial Institute of Technology
State	Telangana
Programme / Specialization	B.Tech, 2nd Year, CSE
Team Name (SIH)	Team ASHTOJ
Email ID	god.s.m.o.97@gmail.com

2. PROJECT INFORMATION

SIH Problem Statement Code	SIH25161
Problem Statement Title	Mitigating National Security Risks Posed by Large Language Models (LLMs) in AI-Driven Malign Information Operations

3. MENTOR / SPOC DETAILS

Assigned Mentor Name	Shri Ojas Dedhiya
Mentor / SPOC Email	Ojasdedhiya@gmail.com

4. PROJECT IMPLEMENTATION SUMMARY

A. Implementation Progress Summary

This week's focus shifted from documentation to internship-level planning — converting last week's literature review into a structured, metric-driven roadmap for the remainder of the engagement. A dedicated planning session was held to define success criteria, scope the problem statement more precisely, and lay out a clear sequence of goals for the coming weeks.

The session produced a working definition of the project's 5W1H (What, Why, Who, When, Where, How), a refined problem statement aligned to the original SIH goal, and a phased roadmap

connecting research depth to eventual prototype hardening. Alongside this, outreach was initiated toward identifying and sourcing additional academic literature, including direct contact with a few researchers to better understand how to locate and evaluate relevant papers.

B. Development & Integration Details

No code-level changes were made this week; the priority was establishing a sound research and planning foundation before further hardening work on TattvaDrishti's pipeline (Ollama semantic scoring, HuggingFace classifier, behavioural and stylometric modules) resumes.

The roadmap drafted this week explicitly sequences the upcoming technical work: deepening the literature base, finalising the problem statement and success metrics, and only then proceeding to enterprise hardening and SOC integration tasks identified in the Week 1 report.

C. Weekly Technical Update

Week 2 (09 June – 14 June 2026):

- Held a structured planning session to define project goals, success metrics, and a working roadmap for the internship
- Drafted a refined problem statement aligned with the original SIH goal
- Defined the solution's 5W1H to clarify scope and direction
- Set a target of reviewing 25–30 literature review papers, with 10–15 to be read in depth; began identifying and downloading candidate papers
- Began outreach to researchers and peers (target: 10–15 contacts) to learn how to source and evaluate relevant literature; initial contact made with a few individuals
- Outlined estimates for the internship's remaining work — split across research, coding, and testing phases — to inform the next two weeks of execution

5. TECHNICAL CHALLENGES / CONSTRAINTS

A. Technical Challenges / Constraints

- Sourcing high-quality, relevant papers within the target range (25–30) is proving time-intensive, particularly for niche sub-topics like federated threat intelligence sharing
- Determining the right balance between research depth and development time was a key planning challenge this week, since over-indexing on literature could delay the hardening phase

B. Operational / Integration Issues

- Outreach to researchers/PhDs for guidance on literature sourcing is still in early stages; response rate so far has been limited
- NDA and formal onboarding documentation from AICTE are still pending, which continues to limit work to the prototype level rather than AICTE infrastructure

6. PLANNED TECHNICAL ACTIVITIES & NEXT STEPS

A. Planned Technical Activities

- Continue reviewing the targeted 25–30 papers, with focus on reading 10–15 in depth over the coming week
- Finalise the problem statement and success metrics drafted in this week's planning session
- Continue outreach to researchers/PhDs (target: 10–15 total contacts) for guidance on the literature review process
- Translate the roadmap's first phase into concrete technical tasks for resuming TattvaDrishti development

B. Upcoming Development Tasks

- Begin enterprise hardening of the TattvaDrishti backend per the roadmap (carried over from Week 1: authentication, input validation, logging)
- Reassess SQLite-to-PostgreSQL migration timeline based on finalised roadmap
- Revisit the SOC integration proposal in light of the refined problem statement and goals

7. IMPLEMENTATION STATUS

Current Status:

- ☐ Not Started
- ☒ In Progress
- ☐ Completed
- ☐ Blocked

Attendance Status:

- ☒ Regular
- ☐ Irregular
- ☐ On Leave

Reporting Week No.

Week 2

Date of Submission	15 June 2026
--------------------	--------------

8. MENTOR / SPOC OBSERVATION & REMARKS

--

Mentor / SPOC Signature	
Date	

Feasibility analysis

TattvaDrishti

Feasibility Analysis of Current Pipeline and Possibility of Multi-Platform Batch Ingestion

Team ASHTOJ | KMIT, Hyderabad | SIH25161 | AICTE Cyber Security Internship 2025–2026

Prepared by	Shaik Mohammed Omar, Shoaib Sadiq Salehmoahmed
Mentor	Shri Ojas Dedhiya
Problem Statement	SIH25161 : Mitigating National Security Risks Posed by LLMs in AI-Driven Malign Information Operations
Document Type	Technical Feasibility Analysis
Date	June 2026

1. Purpose

This document presents a structured feasibility analysis of TattvaDrishti's current detection pipeline and evaluates the viability of extending the system to ingest and analyse content from multiple social media platforms (Twitter/X, Reddit, Telegram, and Facebook) in a batch processing mode. The analysis is intended to inform the next phase of internship development and to give the assigned mentor a clear picture of what is technically achievable, what the key constraints are, and what the recommended path forward looks like.

2. Current Pipeline Overview

TattvaDrishti is a FastAPI-based backend that processes submitted text content through a four-signal detection pipeline and stores results in a SQLite database. A Next.js analyst dashboard consumes the results via REST endpoints and a real-time Server-Sent Events (SSE) stream.

2.1 Detection Pipeline — Signal Weights

- **40%** — Ollama Local LLM Semantic Scoring
- **35%** — HuggingFace RoBERTa Classifier (roberta-base-openai-detector)
- **15%** — Behavioural Analysis (urgency, manipulation tactics, authority claims)
- **10%** — Stylometric Analysis (vocabulary richness, sentence-level features)

2.2 Supporting Infrastructure

- Ingestion: POST /api/v1/intake — single-item JSON payload
- Persistence: SQLite at data/app.db (synchronous writes)
- Sharing: Fernet encryption + Ed25519 signatures for federated packages
- Real-time: SSE stream at /api/v1/events/stream
- Frontend: Next.js analyst dashboard with case drill-down and graph visualisation

2.3 What the Pipeline Does Not Currently Have

The prototype has no social media ingestion layer. There is no scraping module, no API integration with any platform, and no queue or scheduler for processing content at volume. All content currently reaches the pipeline through a manual form submission on the dashboard or a direct API call. This is the primary gap that batching would address.

3. Feasibility Analysis — Current Pipeline

Before adding batch ingestion, it is important to assess how well the existing pipeline would hold up under increased load. The table below breaks down each component's current behaviour, how it would behave in batch mode, where the bottleneck lies, and the estimated development effort to make it batch-capable.

Pipeline Component	Current Mode	Batch Mode	Bottleneck	Dev Effort
FastAPI Ingestion Endpoint	Single-item POST /intake	POST /intake/batch (list payload)	Minimal	Low
HuggingFace RoBERTa Classifier	Single text inference	Native batch inference (tokenizer supports lists)	GPU/CPU memory	Low
Ollama Semantic Scorer	Single prompt, one LLM call	Sequential or queue-based; no native batching	High — LLM is the slowest stage	High
Behavioural Analysis	Per-item rule scoring	Fully parallelisable (stateless)	Negligible	Low
Stylometric Engine	Per-item feature extraction	Fully parallelisable (stateless)	Negligible	Low
SQLite Persistence	Synchronous single writes	Bulk INSERT supported; WAL mode needed	Concurrent write locks	Medium
SSE Event Stream	Per-case broadcast	Batch-complete event or progress events	Negligible	Low

3.1 Critical Bottleneck: Ollama Semantic Scoring

The Ollama integration is the single most significant constraint on batch feasibility. Unlike the HuggingFace classifier, which supports native batched tokenisation and inference, Ollama processes one prompt at a time. Each LLM call for semantic scoring takes approximately 3–15 seconds depending on hardware and model size (llama3.2:3b on CPU). For a batch of 50 social media posts, this translates to 2.5–12.5 minutes of sequential LLM inference before results are available.

Two mitigation strategies exist: (a) run Ollama calls asynchronously using a task queue (Celery + Redis, or Python asyncio with a semaphore-limited worker pool), which allows other pipeline stages to run in parallel; or (b) reduce Ollama's weight in the batch pipeline and rely more heavily on the faster

HuggingFace classifier for high-throughput scenarios, reserving the full LLM score for flagged high-risk items only.

3.2 SQLite Concurrency Limitation

SQLite's default journal mode does not support concurrent writes efficiently. Under batch processing where multiple results are written simultaneously, write lock contention will cause slowdowns or errors. The fix is enabling Write-Ahead Logging (WAL) mode (`PRAGMA journal_mode=WAL`), which is a single-line configuration change and is fully compatible with the existing SQLAlchemy setup. For larger batches (>500 items), migration to PostgreSQL should be planned.

3.3 Components That Are Already Batch-Ready

The behavioural analysis and stylometric modules are stateless and deterministic — they apply rule-based scoring to individual text inputs with no shared state. These can be parallelised trivially using Python's `concurrent.futures.ThreadPoolExecutor` or `asyncio.gather`. The HuggingFace tokeniser and model also support batched list inputs natively, making the classifier stage straightforward to adapt.

4. Feasibility Analysis — Social Media Platform Ingestion

The following analysis covers each of the four target platforms: Twitter/X, Reddit, Telegram, and Facebook. For each platform, the key questions are: does a usable API exist, what content can be retrieved, and what are the practical constraints.

Platform	API Availability	Content Volume	Content Format	Key Challenge	Feasible
Twitter / X	Restricted (Basic tier: 500k reads/mo)	Medium–High	Text, URLs, threads	API access cost, rate limits	Partial
Reddit	Open (PRAW / Reddit API)	Medium	Posts, comments, threads	Nested thread structure, subreddit scope	Yes
Telegram	Telegram Bot API / MTProto	High (public channels)	Text, media, forwarded msgs	No central search; channel discovery required	Partial
Facebook	Meta Graph API (very restricted)	Low–Medium	Posts, pages, groups	Group/page access heavily gated post-2018	Limited

4.1 Twitter / X

Twitter's API has undergone significant restrictions since 2023. The free tier no longer supports read access; the Basic tier (USD 100/month) allows 500,000 tweet reads per month, which is sufficient for a research-grade prototype but not for production-scale monitoring. The most feasible approach is keyword- and hashtag-based search using the v2 search endpoint, which returns recent tweets matching a query.

Streaming endpoints (filtered stream) are available on the Basic tier and allow real-time ingestion of matching tweets — this is the closest existing capability to true batch ingestion from the platform.

4.2 Reddit

Reddit provides the most accessible API for this use case. The PRAW (Python Reddit API Wrapper) library offers clean, well-documented access to posts, comments, and subreddit feeds. Reddit's API allows up to 60 requests per minute (OAuth), and each request can return up to 100 posts. Targeted subreddits (e.g., *r/worldnews*, *r/conspiracy*, *r/india* for domestic threat monitoring) can be polled on a schedule. Reddit is the recommended starting platform for a batch ingestion proof-of-concept due to its permissive API access.

4.3 Telegram

Telegram presents a unique profile. Public channels can be accessed via the official Bot API (polling or webhook), but channel discovery must be done manually — there is no Telegram-native search for channels by topic. The Telethon library (MTPROTO client) provides more powerful access including message history retrieval, but requires a verified phone number for account creation. For high-risk disinformation monitoring, Telegram public channels are a high-value source and integration is feasible with moderate effort, though channel curation is a manual step.

4.4 Facebook

Facebook is the most constrained of the four platforms. Following the Cambridge Analytica incident, Meta locked down the Graph API substantially. Access to public page content is still available, but group content requires Group admin approval, and user content is inaccessible. CrowdTangle, previously used by researchers for Facebook and Instagram monitoring, was shut down in August 2024. For the current internship scope, Facebook integration is marked as low-feasibility and is recommended for a later phase.

5. Proposed Batch Architecture

Based on the above analysis, the following architecture is recommended for adding multi-platform batch ingestion to TattvaDrishti.

Layer 1 — Platform Connectors (new module: `app/connectors/`)

- `RedditConnector`: PRAW-based poller, configurable subreddit list and polling interval
- `TwitterConnector`: Tweepy v2 client, keyword/hashtag search or filtered stream
- `TelegramConnector`: Telethon or Bot API webhook, curated channel list
- `FacebookConnector`: Graph API page scraper (limited scope, low priority)
- Each connector normalises output to a standard `ContentItem` schema before passing to the pipeline

Layer 2 — Ingestion Queue (new: `Redis + Celery` or `asyncio` queue)

- Connectors push `ContentItems` into an in-memory or Redis-backed queue
- A configurable batch size (e.g., 20–50 items) triggers a batch analysis run
- Alternatively, a scheduled cron trigger (e.g., every 5 minutes) drains the queue

Layer 3 — Adapted Pipeline (modifications to existing app/services/orchestrator.py)

- HuggingFace classifier: switch single-item call to batched list inference
- Behavioural + stylometric modules: wrap in `asyncio.gather` for parallel execution
- Ollama scorer: run via a semaphore-limited async worker pool (max 2–3 concurrent calls); for very large batches, apply only to items flagged high-risk by the classifier
- SQLite: enable WAL mode; use bulk INSERT for batch results

Layer 4 — Dashboard Updates (minor frontend changes)

- Add a batch status indicator to the analyst dashboard
- SSE stream to emit progress events during batch processing (e.g., '12/50 items analysed')

6. Identified Gaps and Risks

6.1 No Social Media Ingestion Layer (Current Gap)

As noted in Section 2.3, the prototype has no connectors, no queue, and no scheduler. This is the largest single gap between the current prototype and the proposed batch-capable system. Building even one connector (Reddit recommended as the first) and a minimal queue would represent a meaningful step toward multi-platform coverage.

6.2 Ollama Throughput Risk

If Ollama is run synchronously on batches of 50+ items, processing time will exceed practical limits for near-real-time use cases. The async worker pool mitigation is necessary before batch mode is usable at any meaningful scale.

6.3 API Cost and Rate Limits

Twitter/X API access requires a paid subscription. Reddit and Telegram are free but rate-limited. The prototype should implement exponential backoff and rate-limit-aware polling in each connector to avoid API bans.

6.4 Content Normalisation Complexity

Each platform has a different content structure — Reddit posts have titles, bodies, and nested comments; Telegram messages may contain forwarded metadata; tweets have character limits and thread structure. A normalisation layer that extracts clean text, source metadata, and timestamp from each format is a non-trivial engineering task and must be accounted for in the roadmap.

6.5 SQLite Scalability Ceiling

SQLite is appropriate for the prototype and small-scale batch runs. For continuous multi-platform ingestion at production volume, a migration to PostgreSQL is necessary. This migration is low-risk (SQLAlchemy abstracts the database layer) but requires provisioning a PostgreSQL instance in the deployment environment.

7. Recommendations

The following phased approach is recommended:

Phase 1 — Batch-Ready Pipeline (2–3 weeks)

- Enable HuggingFace batched inference in orchestrator.py
- Implement asyncio-based parallel execution for behavioural and stylometric modules
- Enable SQLite WAL mode and implement bulk INSERT for batch results
- Add a POST /api/v1/intake/batch endpoint accepting a list of content items

Phase 2 — Reddit Connector + Queue (2 weeks)

- Build RedditConnector using PRAW targeting 3–5 high-risk subreddits
- Implement a lightweight asyncio queue to buffer connector output
- Wire connector output to the batch endpoint with a configurable trigger

Phase 3 — Ollama Async Worker Pool (1–2 weeks)

- Wrap Ollama calls in a semaphore-limited async pool (max 2–3 concurrent)
- Implement a fast-path option: skip Ollama for items scoring below 0.3 on HuggingFace

Phase 4 — Twitter/X and Telegram Connectors (2–3 weeks)

- Add TwitterConnector using Tweepy v2 filtered stream or keyword search
- Add TelegramConnector via Telethon for curated channel monitoring
- Facebook deferred to a later phase given API access constraints

8. Conclusion

TattvaDrishti's current pipeline is well-architected for extension to batch processing. The HuggingFace classifier, behavioural, and stylometric modules are already batch-compatible with minimal changes. The Ollama integration is the primary performance constraint, addressable through an async worker pool. SQLite requires WAL mode enabled before batch writes are reliable.

Of the four target platforms, Reddit is the most immediately feasible for batch ingestion and is recommended as the starting point. Twitter/X is feasible with a paid API tier. Telegram is feasible for public channels with moderate setup effort. Facebook remains the most constrained and is low-priority for the current internship scope.

The proposed four-phase roadmap is achievable within the internship timeline. Building even Phase 1 and Phase 2 (batch-ready pipeline + Reddit connector) would transform TattvaDrishti from a manual single-item prototype into a functioning automated multi-source detection system — a meaningful step toward enterprise deployment within AICTE's cyber security infrastructure.

Script

TattvaDrishti didn't start as a research project or a company initiative—it started as an idea for the Smart India Hackathon 2025. For those unfamiliar, the Smart India Hackathon is India's largest nationwide innovation competition, where students develop technology solutions to address real-world problems identified by government ministries and industry. The team developed this solution during SIH, where it went on to win the competition. Following that achievement, they were selected for the AICTE Cyber Security Internship, where they have been working to transform that winning prototype into a much more capable platform. What you'll see today is the result of that journey—from a hackathon concept to an evolving analyst-centric intelligence platform."

Shoaib: slide-2

Good morning, respected members of the panel. I am Shoaib Sadiq Salehmohamed, and presenting alongside me is my teammate, Shaik Mohammed Omar. and today we Present to you TattvaDrishti - an analyst-centric intelligence platform for AI-enabled malign information operations. ~~TattvaDrishti began as our Smart India Hackathon 2025 winning solution and is now being developed further through the AICTE Cyber Security Internship.~~ Our objective is simple: to help authorized analysts identify, understand and investigate coordinated disinformation faster, without replacing human judgement.

:Every solution begins with understanding the problem. Let's take a moment to see why a platform like TattvaDrishti has become more important than ever.

[Video]

Omar: Slide-3

Today, analysts in cybersecurity teams, police departments and intelligence agencies often rely on multiple disconnected tools to investigate a single incident. Platforms like LangSmith and Galileo help **evaluate AI responses**, Maltego helps uncover **network relationships**, OpenCTI supports **cyber threat intelligence**, while **fact-checking** and **social listening platforms** each solve

their own part of the problem. Each provides valuable insights but only into one piece of the investigation.

()

That's exactly the gap TattvaDrishti is designed to address. Investigators spend valuable time moving between different platforms instead of focusing on the investigation. TattvaDrishti brings the **entire workflow** into one place, making it easier to collect information, understand it, uncover hidden patterns and securely share the findings.

Omar: Slide-4

Today, our prototype supports both individual and batch intake and supports importing content from social media platforms like Telegram and Whatsapp.

It then analyzes the information, highlights suspicious content, connects related messages, and helps investigators understand how information is spreading, all through one simple, connected platform.

Shoaib: Slide-5

Our main differentiation is that we do not depend on one opaque detector. We retain multiple independent evidence signals: semantic risk, possible AI origin, behavioural manipulation indicators, stylometric patterns, actor relationships, repeated narratives and provenance information. These signals support an explainable analyst assessment. A high-risk score means that an item deserves attention; it does not independently determine truth, guilt, intent or enforcement.

We will now demonstrate how this works on the current prototype.

[

Live Demo:

Shoaib: Demo-1

This is the current TattvaDrishti analyst dashboard running locally on a laptop. For this demonstration, we are using a controlled dataset containing routine conversation alongside synthetic and publicly sourced misinformation patterns related to a developing public event. We are not classifying lawful protest or public discourse as malicious; we are examining attempts to exploit such an event through false claims and coordinated amplification.

[This is a hypothetical message that falsely claims that a bank is about to collapse and pressures people to immediately withdraw their money. Similar panic-driven situations, such as those seen during the Silicon Valley Bank crisis, show how fear and rumours can quickly influence public behaviour.]

]

Omar: Demo-2 (Manual enter the message)

An analyst can submit the text together with available context such as source, platform, region, actor identifier, language and observation time. That context matters because the same sentence can have very different meanings in different situations. The content then passes through several independent analytical modules rather than one classifier.

[Enter the message and hit run analysis]

(Result Table)

: Every analysis is automatically saved in the current session. Here, the analyst can quickly see all previously analyzed messages, their priority level, overall risk score and the AI model used. This makes it easy to revisit any case at any time.

(Case Intelligence)

: the analyst receives a quick summary of the case, including the overall priority level, the composite score and the basic case information. This gives an immediate understanding of how important the message may be.

(Signal breakdown)

: Below that, the platform breaks the analysis into different categories, such as writing style, behavioural indicators and the likelihood of AI-generated content. This helps the analyst understand which factors contributed most to the final assessment.

(Model family)

: The system also estimates which **model family** the content most closely resembles. This is supporting evidence that may help during an investigation.

(Stylometric analysis)

: Here we can see the writing pattern of the message. Rather than expecting the analyst to interpret technical measurements, TattvaDrishti uses these patterns internally to identify unusual or repetitive writing behaviour that may deserve closer attention.

(Provenance)

: Where possible, TattvaDrishti also checks the origin and integrity of the content, helping analysts understand whether supporting information is available for further verification.

: Not every AI-generated message is harmful, and not every harmful message is created by AI. That's why TattvaDrishti looks at multiple signals instead of relying on just one. The platform provides the evidence.

[]tracing back to the original user or the source[]

Shoaib: Demo-3 (BATCH UPLOAD)

Investigations rarely contain only one message. Analysts may receive hundreds or thousands of records from an authorized export or monitored public source. TattvaDrishti therefore supports structured batch intake. The system validates every record, isolates malformed entries, preserves original identifiers and processes valid items together. This demonstration collection contains approximately 150 messages. On our current test setup, a similar collection has been processed in approximately 10 to 15 seconds after batching and parallelization. **This is an internal engineering observation, not yet a formal performance benchmark. While the live job runs, we have the completed analysis of the same collection available here.**

: Instead of reviewing hundreds of messages one by one, TattvaDrishti analyzes them together and presents a clear, organized view of the findings. Similar messages and repeated patterns are grouped together, making it much easier for investigators to understand what's happening and focus on what matters most.

Shoaib: Demo-4 (Threat Graph)

: This is where individual messages become a bigger story. Instead of looking at one message at a time, TattvaDrishti connects related messages, accounts and information to show how they are linked. This helps investigators quickly spot repeated patterns and understand whether multiple messages are part of the same larger activity.

Shoaib: Demo-5 (Actor view & Auditing)

: The platform also lets investigators look at an account in more detail. Instead of judging someone based on a single post, TattvaDrishti provides the complete picture—what they shared, how their activity changed over time, and how it connects to other related content. This helps investigators make informed decisions based on evidence rather than assumptions.

: Once the investigation is complete, the findings can be securely shared with other authorized teams. The important evidence is preserved, sensitive information can be protected, and every step is recorded for transparency. This ensures that everyone works with the same trusted information throughout the investigation.

Omar: Demo-6 (Guided dashboard)

: This is our Guided Mode, designed for officers and personnel who need quick answers without technical complexity. They simply paste a message, image or suspicious content, add basic context if available, and TattvaDrishti analyzes it in seconds. Instead of overwhelming the user with technical details, the platform clearly explains the priority level, highlights the key reasons behind the result, and suggests whether the content requires further attention. This allows officers in the field to make faster, more informed decisions while leaving deeper investigation to specialized analysts when needed.

]

Omar: Slide-8

Every user has a different responsibility. That's why TattvaDrishti provides a simple interface for field officers, a detailed investigation workspace for analysts, and a management dashboard for administrators as we've seen in our demo. Each person sees exactly what they need nothing more, nothing less.

Shoaib: Slide-9

Since TattvaDrishti may be used to analyze sensitive information, security and privacy are a top priority. The platform is designed to protect data, ensure only authorized users can access it, and allow information to be shared safely between trusted teams. Every investigation remains transparent and traceable, while human judgement always stays at the center of the decision-making process.

Shoaib: Slide-10

While TattvaDrishti is already a working prototype, we see this as just the beginning. Our next focus is making the platform faster, more secure, more scalable and capable of supporting multiple Indian languages. Most importantly, we want to work closely with police, cybersecurity experts and intelligence agencies to ensure the platform solves real-world challenges and makes investigations faster, easier and more reliable.

Omar: Conclusion

Misinformation is no longer just a problem on the internet, it has real-world consequences. We believe investigators deserve tools that are faster, smarter and easier to use. TattvaDrishti brings scattered information together, helping turn uncertainty into understanding and information

into actionable intelligence. Most importantly, it keeps the human investigator at the center of every decision.

We sincerely thank you for your time and look forward to your guidance in taking this vision forward. Thank you.

Tab 11

 URGENT: Bharat Vikas Bank (BVB) IN CRISIS 

 CONFIDENTIAL 

My cousin works at BVB headquarters in Mumbai. He just called me in panic. The bank is facing a LIQUIDITY CRISIS and will suspend all withdrawals by 6 PM TODAY.

INSIDER INFORMATION:

Bank has lost ₹8,000 crores in bad loans

RBI emergency meeting happening NOW

Branches ordered to stop cash disbursement by evening

ATMs will be emptied by tonight

THIS IS NOT A DRILL.

My cousin risked his job to warn family. He said "Tell everyone you know to withdraw their savings IMMEDIATELY. Once the announcement comes, it will be too late."

Already 200+ account holders are queuing at the Lokhandwala branch. Cash is running out fast.

 ACT NOW BEFORE 6 PM 

If you have money in Bharat Vikas Bank:

Rush to your nearest branch NOW

Withdraw maximum possible amount

Tell your friends and family immediately

Forward this to everyone you know who banks with BVB. Don't let them lose their life savings.

A friend of my colleague already lost ₹15 lakhs in a similar situation last year. Don't wait for official confirmation - by then your money will be GONE.

● SHARE THIS IN ALL YOUR GROUPS ●